

الورقة الامتحانية
لامتحان الشهادة الثانوية الأزهرية - نموذج تجريبي
للعام الدراسي ١٤٤٧ هـ - ٢٠٢٥ / ٢٠٢٦ م

المادة/الكيمياء (لغات) (تجريبي)	
القسم العلمي	زمن الإجابة: ثلاث ساعات

السؤال	الدرجة بالأرقام	اسم المصحح ثلاثيًا
الأول		
الثاني		
الثالث		
الرابع		
الخامس		
المجموع		

عدد الصفحات (١٦) صفحة
و على الطالب مسئولية المراجعة
والتأكد من ذلك قبل تسليم الكراسة

الرقم السري

مجموع الدرجات (مكتوباً بالحروف):

اسم المراجع العددي ثلاثيًا: اسم المراجع الفني ثلاثيًا:

الأزهر الشريف - قطاع المعاهد الأزهرية - الإدارة المركزية للامتحانات وشؤون الطلاب و الخريجين

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الرقم السري

نموذج ثانوية الأزهرية تجريبي

اسم الطالب كاملاً:
رقم الجلوس: اسم المعهد:
التاريخ: / /

تعليمات هامة

عزيزي الطالب / عزيزتي الطالبة:

- اقرأ السؤال بعناية، وفكر فيه جيداً قبل البدء في الإجابة عليه.
 - أجب عن جميع الأسئلة ولا تترك أي سؤال دون إجابة.
 - عند إجابتك عن الأسئلة المقالية، أجب فيما لا يزيد عن المساحة المحددة لكل سؤال.
 - عند إجابتك عن أسئلة الاختيار من متعدد (إن وجدت)، ظلل الدائرة ذات الرمز الدال على الإجابة الصحيحة فقط تظليلاً كاملاً.
- مثال: الإجابة الصحيحة (د) مثلاً

أ	ب	ج	د
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- في حالة التظليل على أكثر من رمز، تعتبر الإجابة خطأ.
 - في حالة ما إذا أجبنا إجابة خطأ، ثم قمنا بشطبها وأجبنا إجابة صحيحة تُحسب الإجابة صحيحة.
 - في حالة ما إذا أجبنا إجابة صحيحة، ثم قمنا بشطبها وأجبنا إجابة خطأ تُحسب الإجابة خطأ.
- ملحوظة:** يفضل عدم تكرار الإجابة على الأسئلة.
- عدد صفحات الكراسة (١٦) صفحة.
 - تأكد من عدد صفحات كراستك، فهي مسئوليتك.
 - زمن الامتحان (ثلاث ساعات).
 - الدرجة الكلية للامتحان (٦٠) درجة.
 - عدد الأسئلة (٥) لكل سؤال (١٢) درجة ، لكل جزئية (درجة واحدة)
 - عند احتياج الطالب للإجابة على أي فقرة وذلك عند حدوث أي سبب يقتضي ذلك؛ يستخدم المسودة بآخر الورقة الامتحانية مع كتابة رقم السؤال والفقرة بوضوح، بشرط ألا تكون الإجابة مكررة.

هذا الجزء

غير مخصص للإجابة

Answer the following questions: (12 Marks for each)

[First question]

(A) Choose the correct answer for each of the following:

1- Which of these methods is best for obtaining acetaldehyde?

- (a) Oxidation of ethanol (b) Reduction of acetic acid
(c) Catalytic hydration of acetylene (d) All of the previous

2- When H_2S gas is passed through a copper (II) sulfate solution:

- (a) A black precipitate forms (b) A black precipitate forms after heating
(c) A white precipitate forms (d) No precipitate forms

3- In the following equilibrium system:



what happens when drops of HCl are added to this system?

- (a) It reacts to the left and $[\text{CH}_3\text{COO}^{-}]$ increases
(b) It reacts to the left and $[\text{CH}_3\text{COO}^{-}]$ decreases
(c) It reacts to the right and $[\text{CH}_3\text{COO}^{-}]$ increases
(d) It reacts to the right and $[\text{CH}_3\text{COO}^{-}]$ decreases

4- One of these reactions is not suitable for preparing iron (II) sulfate:

- (a) reaction of iron with dilute sulfuric acid
(b) reaction of iron (II) oxide with dilute sulfuric acid
(c) reaction of iron with copper sulfate II
(d) reaction of iron oxide III with concentrated sulfuric acid

5 - We can distinguish between ethanol and ethanoic acid using

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- (a) sodium (b) potassium
(c) sodium bicarbonate (d) all are correct

6- Many chemical reactions approximately double in speed if the temperature increases by degrees Celsius.

- (a) 10°C (b) 20°C (c) 30°C (d) 40°C

(B) If the mass of a sample of hydrated iron (III) oxide is 1.870 g and it is heated intensely until its mass is constant and found to be 1.60 g, calculate the percentage of water of crystallization in the sample. Then, find the number of water of crystallization molecules and the molecular formula of the oxide (O = 16, H = 1, Fe = 56).

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(C) Three compounds (A, B, C) contain a hydroxyl group.

- a) A reacts with sodium but does not react with sodium hydroxide.
- b) B reacts with both sodium and sodium hydroxide.
- c) C its solution gives with one of the solutions of iron (III) salts, a reddish-brown precipitate.

1- Write the names of the three compounds mentioned

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2- From compound (A), how can you obtain a compound with pungent Odour.

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3- From compound (B), how can you obtain a compound used in the treatment of burns?

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[Second question]**(A) Choose the correct answer for each of the following:**

- 1- The product of the catalytic hydration of propyne is

☐ (a) propanone	☐ (b) propanol
☐ (c) tertiary alcohol	☐ (d) both a and b
- 2- When barium chloride solution is added to a solution of salt, a white precipitate is formed that is insoluble in acids.

☐ (a) nitrate	☐ (b) phosphate
☐ (c) sulfate	☐ (d) sulfide
- 3- When HCl is added to sodium bicarbonate salt, the number of moles of CO₂ released is

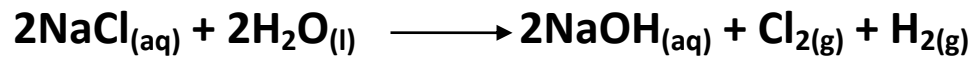
☐ (a) twice what occurs when carbonates are added
☐ (b) half what occurs when carbonates are added
☐ (c) equal in both cases
☐ (d) both a and b are correct
- 4- The amount of electricity required to separate 3 g of aluminum (Al =27) by the electrolysis of molten bauxite is

☐ (a) 1F	☐ (b) 1/3F	☐ (c) 3F	☐ (d) 6F
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- 5- A yellow iron ore that is easily reduced is

☐ (a) hematite	☐ (b) magnetite
☐ (c) siderite	☐ (d) limonite
- 6- The OH group in alcohols

☐ (a) Non-ionized and chemically inactive
☐ (b) Non-ionized and chemically active
☐ (c) Ionized and chemically inactive
☐ (d) Ionized and chemically active

(B) During the electrolysis of sodium chloride solution, chlorine and hydrogen gases are released at the electrodes according to the equation:



1- What is the name of the gas released at each electrode?

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2- Calculate the volume of chlorine gas ($\text{Cl} = 35.5$) released (at STP) when an electric current of 2 A is passed for 20 minutes.

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(C) Rearrange the following steps to obtain nitrobenzene from toluene, with writing Equations:

- 1- Dry distillation in the presence of soda lime
- 2- Oxidation
- 3- Nitration
- 4- Reaction with caustic soda solution

Order:

- 1) 2)
- 3) 4)

Equations:

- 1)
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- 2)
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- 3)
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- 4)
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[Third question]**(A) Choose the correct answer for each of the following:**

1- Which of these compounds has the highest boiling point?

- (a) C_2H_5OH (b) CH_3COOH
(c) $HCOOCH_3$ (d) C_2H_6

2- The law of action of mass can be applied to a solution of

- (a) sodium chloride (b) hydrocyanic acid
(c) potassium hydroxide (d) hydrochloric acid

3- To obtain formic acid from methane, the following processes take place in sequence:

- (a) Halogenation $\rightarrow$ Oxidation $\rightarrow$ Hydrolysis
(b) Hydrogenation $\rightarrow$ Hydrolysis $\rightarrow$ Oxidation
(c) Hydrolysis $\rightarrow$ Oxidation $\rightarrow$ Neutralization
(d) Halogenation $\rightarrow$ Hydrolysis $\rightarrow$ Oxidation

4- When iodine solution is added to salt (A), the brown color of the iodine disappears, and the solution of salt (A) is

- (a) potassium sulfate (b) sodium thiosulfate
(c) copper (II) sulfide (d) sodium nitrite

5- Element (X) is used as a catalyst in ammonia is obtained industrially using the Haber-Bosch process. Element (Y) is used in the manufacture of car tires. When the two elements combine chemically, the alloy forms

- (a) duralumin (b) interstitial
(c) cementite (d) substitutional

6- The reduction process takes place in the fuel cell of

- (a) $H_{2(g)}$ (b) $H_2O_{(l)}$
(c) $OH^-_{(aq)}$ (d) $O_{2(g)}$

(B) Calculate the ionization constant and degree of ionization of 0.1 M acetic acid, if its pH value is 3.

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(C) Upon catalytic hydration of compound (X), compound (Y) is formed. Upon oxidation of compound (Y), compound (Z) is formed. Upon catalytic hydration of compound (A), compound (B) is formed. Upon reaction of (B) with (Z), ethyl ethanoate is formed.

Answer the following:

1- What is the chemical formula of compounds (A) and (X)?

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2- What is the name of the chemical process required to obtain (B) from (Y)?

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[Fourth question]

(A) Choose the correct answer for each of the following:

1- FeCl_3 solution gives a purple color with

- (a) carbolic acid (b) salicylic acid
(c) phthalic acid (d) a and b

2- Iron (II) chloride is prepared by reacting iron with

- (a) chlorine gas
(b) dilute hydrochloric acid
(c) air then concentrated hydrochloric acid
(d) all of the previous

3- The following table represents the reduction potentials of different elements. Which of these elements can be used as a sacrificing electrode for the other element?

Element	A	B	C	D
Reduction potential	-1.02	- 2.3	1.02	- 0.12

- (a) D relative to B (b) A relative to C
(c) D relative to A (d) C relative to D

4- To obtain ethane from glucose, the following processes occur in sequence

- (a) Fermentation - Catalytic Hydration - Hydrogenation
(b) Hydrogenation - Fermentation - Catalytic Hydration
(c) Oxidation - Hydrogenation - Fermentation
(d) Fermentation - Dehydration - Hydrogenation

5- The graph representing the rate of a chemical reaction has on the horizontal axis and on the vertical axis.

- (a) rate of reaction / time (b) time / mass
(c) time / concentration (d) time / volume of gas released

6- A solution with a hydrogen ion concentration of 10^{-11} mol/L has a pOH value of

- (a) 10^{-11} (b) 14
(c) 3 (d) 11

(B) State one use for each of the following:

1) Titanium dioxide

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2) Hydrometer

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3) Zinc oxide

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(C) A, B, and C are three transition elements located in the fourth period:

(A) An element that reacts with nitric acid but does not react with hydrochloric or sulfuric acids.

(B) An element with less mass than the element preceding it and less mass than the element following it in the period.

(C) An element that is highly chemically reactive but resistant to Atmospheric air.

From the previous information, answer the following:

1) An element whose one compound are used in the detection of glucose.

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2) Arrange the divalent ions of these elements according to their magnetic moment.

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[Fifth question]

(A) Correct the underlined words in each of the following statements:

- 1- When two equal volumes of sodium hydroxide and sulfuric acid solutions, each with a concentration of 0.5 M, are mixed, the resulting solution is neutral.
- 2- When a ketone is reduced, a tertiary alcohol is formed.
- 3- The first analytical cations group are precipitated as hydroxides.
- 4- In a Daniell cell, the oxidizing agent is Zn^{2+} .
- 5- Alkyl is the radical produced after the removal of a hydrogen atom from ethylene.
- 6- Oils and fats are natural polymers of amino acids.

(B) Write the scientific term for each of the following statements:

- 1) The process of converting normal heptane to toluene.
(.....)
- 2) A polymer produced from the condensation of terephthalic acid with ethylene glycol.
(.....)
- 3) The amount of electricity required to separate the gram equivalent mass of any substance.
(.....)

(C) Give reason for the following:

1) Water is not added to alkenes except in the presence of acidic medium.

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2) The battery must be recharged after a period of operation.

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3) Most industrial metals corrode when mixed with impurities.

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This image shows a full page of a worksheet designed for handwriting practice. It consists of multiple rows of horizontal dashed lines spaced evenly across the page, providing a guide for letter height and placement. The background is plain white, and there are no other markings or text present.

Draft

[illegible]

This image shows a full page of primary-ruled paper. It features multiple sets of horizontal dashed lines spaced evenly down the page, providing a guide for handwriting practice. The background is white, and there are no margins or other markings present.